

GAUSSIAN MULTI-SET CONCENTRATION FROM ISOPERIMETRIC CONTRACTION

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STATUS

This is a candidate full answer, subject to human review, to the Gaussian multi-set concentration conjecture in Section 5.1 of Gozlan and Herry [2]. It gives the requested profile with the stronger constant $C_{k,\rho,n} = \rho/2$, independent of k and n .

SOURCE QUESTION

For $k \geq 1$, let Δ_k consist of all $a = (a_1, \dots, a_k) \in [0, 1]^k$ such that

$$\sum_{j=1}^k a_j \leq 1, \quad a_i + \sum_{j=1}^k a_j \geq 1 \quad (1 \leq i \leq k).$$

A probability metric space has the multi-set concentration property of order k with profile α_k if, whenever the Borel sets $A_1, \dots, A_k$ are pairwise positively separated and $(\mu(A_1), \dots, \mu(A_k)) \in \Delta_k$, their union A satisfies

$$(1) \quad \mu(A_r) \geq 1 - (1 - \mu(A))\alpha_k(r), \quad 0 < r \leq \frac{1}{2} \min_{i \neq j} d(A_i, A_j).$$

Here $A_r = \{x : d(x, A) < r\}$.

The source asks whether every probability measure $d\mu = e^{-V}dx$ on $\mathbb{R}^n$ with V smooth and $\text{Hess } V \geq \rho I > 0$ has this property with a Gaussian profile $\exp(-C_{k,\rho,n}r^2)$.

We list open questions related to the multi-set concentration of measure phenomenon.

5.1. **Gaussian multi-set concentration.** Using the terminology introduced in [Section 4](#), [Theorem 1.1](#) and the material exposed in [Section 2](#) tell us that, if μ has a density of the form e^{-V} with respect to Lebesgue measure on $\mathbb{R}^n$ with a smooth function V such that $\text{Hess } V \geq \rho > 0$, then the probability metric space $(\mathbb{R}^n, |\cdot|, \mu)$ satisfies the multi-set concentration of measure property of order k with the concentration profile

$$\alpha_k(r) = \exp\left(-c \min(r^2 \lambda_{\gamma_{n,\rho}}^{(k)}; r \sqrt{\lambda_{\gamma_{n,\rho}}^{(k)}}\right), \quad r \geq 0,$$

where $\lambda_{\gamma_{n,\rho}}^{(k)}$ denotes the k th eigenvalue of the n -dimensional centered Gaussian measure with covariance matrix $\rho^{-1}\text{Id}$. Since the measure μ satisfies the log-Sobolev inequality, it is well known that it satisfies a (classical) Gaussian concentration of measure inequality. Therefore, it is natural to conjecture that μ satisfies a multi-set concentration of measure property of order $k \geq 1$ with a profile of the form

$$\beta_k(r) = \exp\left(-C_{k,\rho,n} r^2\right), \quad r \geq 0,$$

for some constant $C_{k,\rho,n}$ depending solely on its arguments. In addition, it would be interesting to see how usual functional inequalities (Log-Sobolev, transport-entropy, ...) can be modified to catch such a concentration of measure phenomenon.

PROOF INTUITION

The condition defining Δ_k already forces the union of the k sets to have mass at least $k/(k+1)$, and hence at least $1/2$. Strong log-concavity makes μ a 1-Lipschitz image of the Gaussian measure with covariance $\rho^{-1}I$ by Caffarelli's contraction theorem. Gaussian isoperimetry therefore bounds the enlargement of the *union*; no separate higher-order estimate is needed. A one-line shifted-tail calculation supplies exactly the multiplicative factor $1 - \mu(A)$ required in [\(1\)](#).

TWO ELEMENTARY LEMMAS

Let Φ and ϕ denote the distribution and density functions of a standard one-dimensional Gaussian, and let $\bar{\Phi} = 1 - \Phi$.

Lemma 1 (The Δ_k mass threshold). *If $a = (a_1, \dots, a_k) \in \Delta_k$ and $s = \sum_{i=1}^k a_i$, then*

$$s \geq \frac{k}{k+1} \geq \frac{1}{2}.$$

Proof. Put $b = 1 - s$. The defining inequalities give $a_i \geq b$ for every i . Thus $s \geq kb = k(1 - s)$, which is equivalent to the claim. $\square$

Lemma 2 (Multiplicative Gaussian tail shift). *For $z, t \geq 0$,*

$$\bar{\Phi}(z+t) \leq e^{-t^2/2} \bar{\Phi}(z).$$

Proof. Changing variables $u = v + t$ gives

$$\bar{\Phi}(z+t) = \int_z^\infty \phi(v+t) dv = e^{-t^2/2} \int_z^\infty e^{-tv} \phi(v) dv \leq e^{-t^2/2} \bar{\Phi}(z),$$

because $v \geq z \geq 0$. $\square$

FULL ANSWER

Theorem 1. *Let μ be a probability measure on $\mathbb{R}^n$ with smooth density e^{-V} and suppose*

$$\text{Hess } V \geq \rho I$$

for some $\rho > 0$. For every $k \geq 1$, the space $(\mathbb{R}^n, |\cdot|, \mu)$ has the multi-set concentration property of order k with

$$\alpha_k(r) = \exp\left(-\frac{\rho r^2}{2}\right), \quad r \geq 0.$$

In particular, the conjecture of [2, Section 5.1] holds with $C_{k,\rho,n} = \rho/2$.

Proof. Let $\gamma_{n,\rho}$ be the centered Gaussian probability measure on $\mathbb{R}^n$ with covariance $\rho^{-1}I$. Since

$$V(x) - \frac{\rho}{2}|x|^2$$

is convex, Caffarelli's contraction theorem [1] supplies a 1-Lipschitz transport map F with $F_{\#}\gamma_{n,\rho} = \mu$.

We first record the resulting Gaussian isoperimetric comparison. For a Borel set $E \subset \mathbb{R}^n$, put $B = F^{-1}(E)$. The Lipschitz property gives $B_r \subset F^{-1}(E_r)$, so Gaussian isoperimetry yields

$$(2) \quad \mu(E_r) \geq \gamma_{n,\rho}(B_r) \geq \Phi(\Phi^{-1}(\mu(E)) + \sqrt{\rho}r).$$

Now take positively separated Borel sets $A_1, \dots, A_k$ satisfying the hypotheses of the multi-set property and set $A = \bigcup_i A_i$. Positive separation makes the sets disjoint, hence, with $a_i = \mu(A_i)$,

$$s := \mu(A) = \sum_{i=1}^k a_i.$$

Lemma 1 gives $s \geq 1/2$. Therefore $z = \Phi^{-1}(s) \geq 0$. Taking complements in (2), and applying Lemma 2 with $t = \sqrt{\rho}r$, gives

$$\begin{aligned} 1 - \mu(A_r) &\leq \bar{\Phi}(z + \sqrt{\rho}r) \\ &\leq e^{-\rho r^2/2} \bar{\Phi}(z) \\ &= (1 - \mu(A))e^{-\rho r^2/2}. \end{aligned}$$

This is exactly (1) with the asserted profile. In fact the bound holds for every $r \geq 0$, stronger than the separation-limited range required by the definition. $\square$

VERIFICATION NOTES

The proof uses only two standard sharp inputs: Caffarelli contraction from $\gamma_{n,\rho}$ to a measure whose potential is at least as convex, and Gaussian isoperimetry. All remaining steps are displayed. The direction of the neighborhood inclusion is important: if x is within r of $F^{-1}(E)$, then $F(x)$ is within r of E , hence $(F^{-1}E)_r \subset F^{-1}(E_r)$.

The multiplicative prefactor is not discarded. Since $\overline{\Phi}(\Phi^{-1}(s)) = 1 - s$, Lemma 2 produces precisely the complement mass in the source definition. The assumption $s \geq 1/2$ is exactly what ensures $z \geq 0$ in that lemma.

For a half-space under $\gamma_{n,\rho}$ with mass $1/2$, the exponent $\rho/2$ is asymptotically sharp as $r \rightarrow \infty$, so no larger uniform quadratic coefficient can hold in this formulation without an additional prefactor or linear term.

NOVELTY CHECK AND SCOPE

A bounded search was performed on August 11, 2026. The local run indexes, solution registry, attempts, and parsed-source scan were searched for arXiv:1804.06133 and combinations of “Gaussian multi-set concentration”, “Caffarelli contraction”, and “Gaussian isoperimetry”. External exact-phrase searches found the arXiv and 2020 Potential Analysis versions, the authors’ publication pages, a thesis reproducing the same open question, and later work on adjacent higher-order eigenvalue bounds, but no paper claiming this answer or applying the contraction argument to the exact multi-set profile. Novelty is plausible, not certified; specialist review of MathSciNet and zbMATH remains advisable.

The theorem answers the first question in Section 5.1. It does not answer the separate Section 5.2 converse problem relating multi-set concentration to lower bounds on higher eigenvalues, nor does it construct new modified functional inequalities.

HUMAN REVIEW RECOMMENDATION

Check the convention and scaling in the cited form of Caffarelli’s contraction theorem, then verify the neighborhood inclusion $(F^{-1}A)_r \subset F^{-1}(A_r)$. The rest reduces to Lemmas 1 and 2. If these checks pass, Theorem 1 is a complete and stronger-than-requested solution.

REFERENCES

- [1] L. A. Caffarelli, *Monotonicity properties of optimal transportation and the FKG and related inequalities*, Comm. Math. Phys. **214** (2000), 547–563.
- [2] N. Gozlan and R. Herry, *Multiple sets exponential concentration and higher order eigenvalues*, Potential Anal. **52** (2020), 203–221; arXiv:1804.06133.
- [3] M. Ledoux, *The Concentration of Measure Phenomenon*, Mathematical Surveys and Monographs 89, American Mathematical Society, 2001.